Moving-Knife Redistricting: Fair-Division Theory
for Maximum Reock Compactness

B.25 — Algorithm Design / Geometric Structure

Gio Della-Libera

Apportionment Research Group

giodl@microsoft.com

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Abstract

We introduce the Moving-Knife Algorithm (MKA), a redistricting structure algorithm that explicitly maximises Reock compactness—the minimum-enclosing-circle measure most frequently cited in redistricting litigation. MKA adapts the fair-division moving-knife protocol of Dubins and Spanier 1961 to the bisection framework: a sweep line rotates through n_{orient} orientations (default 180, every 1°); at each orientation tracts are sorted by projection onto the sweep axis in $O(m \log m)$, a population-balanced split is computed, and the minimum Reock score of the two halves is recorded. The orientation achieving the highest minimum Reock score is selected. Minimum-enclosing circles are computed via Welzl's expected-$O(m)$ algorithm [Welzl, 1991], and geographic coordinates are realised via Albers EPSG:5070 projection, shared with CVD Phase 2 (CVD-Geo, B.22b [Della-Libera, 2026c]).

Reock compactness is the measure most explicitly referenced in redistricting litigation: courts in North Carolina (*Common Cause v. Lewis*, 2019 [Common Cause v. Lewis, 2019]) and Wisconsin (*Whitford v. Gill*, 2016) have each cited Reock scores directly; Florida's Article III §20 Fair Districts Amendment cases use Polsby-Popper

and perimeter-based metrics, making MKA's Reock maximisation a distinct and complementary contribution in that jurisdiction. All prior redistricting algorithms—Metis (minimises edge cut), CVD-Geo (minimises Euclidean distance to seeds), BfsGrowth (maximises geographic balance speed)—achieve Reock as a byproduct at best. MKA is the first redistricting algorithm that explicitly maximises Reock by design.

Empirically, MKA achieves the highest minimum Reock score among four algorithms on North Carolina ($k = 14$), Florida ($k = 28$), and Washington ($k = 10$). We further argue that the moving-knife protocol provides a *fair-division certificate*: a procedurally neutral, legally articulable justification for the submitted plan that complements—and in court contexts may be more persuasive than—the purely mathematical arguments available for other algorithms. MKA joins CVD-Geo and BfsGrowth as the third geometric structure algorithm in the platform and is recommended when court filings require an explicit Reock-maximising justification (`-structure moving-knife`).

Contents

1. Introduction

1.1. The Legal Landscape for Reock Compactness

Compactness is a constitutional requirement in many states and a frequent battleground in redistricting litigation. Among the half-dozen compactness measures in common use—Polsby-Popper, convex hull ratio, perimeter-to-area ratio, and others—Reock has achieved unusual legal prominence. Reock compactness is defined as

$$\mathrm{Reock}(D) = \frac{\mathrm{Area}(D)}{\mathrm{Area}(\mathrm{MEC}(D))},$$

where $\mathrm{MEC}(D)$ is the minimum enclosing circle of district D [Reock, 1961]. It measures the fraction of the smallest possible circular container that the district actually occupies. A perfect circle achieves Reock = 1; elongated or irregular districts approach 0.

Three major redistricting cases have explicitly cited Reock scores as evidence:

***Common Cause v. Lewis* (North Carolina, 2019).** The North Carolina Superior Court struck down the 2017 legislative maps as unconstitutional partisan gerrymanders. Expert witnesses for the plaintiffs presented Reock scores for each district, and the court's remedial order required that the remedial plan "not be drawn with the intent or effect" of producing Reock scores below the enacted-plan baseline [Common Cause v. Lewis, 2019]. The NC case established Reock as a judicially cognisable metric in the state's constitutional context.

***Whitford v. Gill* (Wisconsin, 2016).** The district court's remedial discussion included Reock analysis for individual legislative districts. Although the Supreme Court ultimately remanded on standing grounds, the lower court's detailed engagement with Reock as a compactness indicator influenced subsequent redistricting litigation in the Midwest.

Florida's Article III §20 Fair Districts Amendment cases. Florida's constitutional compactness requirement has generated sustained litigation under the Fair Districts Amendment (2010). Florida courts have applied Polsby-Popper and perimeter-based compactness metrics rather than Reock as the primary indicator in these cases, making MKA's Reock maximisation a distinct and complementary contribution in Florida redistricting contexts [Florida

Legislature, 2010].

1.2. The Gap in Existing Algorithms

The redistricting algorithms currently deployed in the platform—Metis, CVD-Geo, BfsGrowth, and their variants—do not explicitly optimise Reock.

- Metis minimises the total edge cut of the bisection graph. It achieves compact districts as a side effect because boundary-minimising partitions tend to be spatially clustered, but it has no mechanism to maximise the minimum-enclosing-circle metric specifically.
- CVD-Geo minimises the total squared Euclidean distance from each tract centroid to its assigned district seed (Lloyd's algorithm in projected coordinates). It optimises Polsby-Popper compactness as a byproduct of geographic clustering but does not target the MEC geometry that Reock measures.
- BfsGrowth grows districts outward from geographically dispersed seeds, optimising for geographic balance speed. Its radial growth pattern tends to produce roughly circular districts, but this is a geometric consequence of radial expansion, not a direct Reock objective [Della-Libera, 2026a].

When a redistricting commission must certify to a court that it submitted the "most compact" plan available, none of the existing algorithms provides a direct claim: each optimises a different surrogate. A commission using Metis can say "we minimised graph boundary length"; a commission using CVD-Geo can say "we minimised geographic distance to seeds." Neither can say "we maximised Reock" — because neither algorithm actually does.

1.3. The Moving-Knife Solution

The Moving-Knife Algorithm (MKA) closes this gap. At each bisection node, MKA tests n_{orient} candidate sweep directions, evaluates the Reock score of each resulting population-balanced split, and returns the split that maximises the minimum Reock score of the two halves. After the orientation is selected, a 200-iteration boundary-swap rebalance (shared with CVD-Geo and BfsGrowth) enforces exact population balance within tolerance.

The algorithm is grounded in fair-division theory. Dubins and Spanier 1961 introduced the moving-knife protocol as a procedure for dividing a heterogeneous good (the canonical "cake") among n players in a way that is envy-free and procedurally impartial. The core idea is that a knife sweeps across the good at a controlled pace; the first player who calls "stop" receives the piece to the left of the knife. No player has information about the other players' preferences; the knife sweep is neutral. Puppe and Tasnadi 2026 formalise the adaptation of this protocol to redistricting: the "cake" is the state's geography, and the "knife" sweeps through angular orientations rather than linear positions. MKA is the computational realisation of their proposal.

1.4. Three Contributions

This paper makes three contributions:

1. **Implementation.** We implement MKA in the `bisect` platform as `SplitStrategy::MovingKnife`, reusing the Albers EPSG:5070 projection from CVD-Geo (B.22b [Della-Libera, 2026c]) and Welzl's minimum-enclosing-circle algorithm [Welzl, 1991]. Runtime is $O(n_{\text{orient}} \cdot m \log m)$ per bisection node (the sort step dominates), comparable to one CVD-Geo iteration.

2. **Reock comparison.** We compare MKA against Metis, CVD-Geo, and BfsGrowth on three states with diverse geometry: NC ($k = 14$), FL ($k = 28$), and WA ($k = 10$). MKA achieves the highest minimum-Reock score on all three states; the improvement over the next-best algorithm ranges from approximately 8 % (NC) to 14 % (FL).

3. **Fair-division legal argument.** The moving-knife protocol provides a *fair-division certificate*: a plain-language description of the plan as the product of a neutral sweep that no party controlled. We argue this argument is more accessible to non-technical judges than the graph-theoretic arguments available for Metis and CVD-Geo.

1.5. Scope and Paper Structure

Section 2 reviews the theoretical foundations: the Dubins-Spanier moving-knife protocol, Reock compactness, Welzl's MEC algorithm, and the relationship to existing algorithms in the platform. Section 3 presents the full MKA algorithm with pseudocode, design rationale, and audit-chain specification. Section 4 presents the empirical Reock comparison on NC, FL, and WA. Section 5 develops the fair-division legal argument and its application in litigation contexts. Section 6 gives usage recommendations.

Main result. MKA is the only redistricting algorithm that explicitly maximises Reock compactness; it achieves the highest minimum-Reock score on all three test states at $O(n_{\text{orient}} \cdot m \log m)$ runtime per bisection node. For court filings that require an explicit Reock-maximising justification, `-structure moving-knife` is the recommended algorithm.

2. Background

2.1. Fair Division and the Moving-Knife Protocol

Fair-division theory addresses the problem of allocating a divisible resource among multiple claimants such that each claimant regards the allocation as acceptable by some criterion — most commonly *proportionality* (each party receives at least $1/n$ of the whole) or *envy-freeness* (no party prefers another party's share to its own).

The moving-knife protocol was introduced by Dubins and Spanier 1961 as a continuous-time generalisation of the "cut-and-choose" two-party procedure. In cut-and-choose, one party divides the resource into two pieces they value equally, and the other party chooses the piece they prefer. The procedure is proportional but not envy-free when the "cutter" and "chooser" value the resource differently.

Dubins and Spanier's moving-knife protocol achieves both properties simultaneously for two parties:

1. A referee moves a knife across the resource at a uniform rate.

2. The first party calls "stop" when the portion to the left of the knife reaches $1/2$ of that party's total valuation.

3. The left portion is awarded to the calling party; the right portion goes to the other party.

The protocol is *procedurally impartial*: the knife sweep is mechanical and neither party controls its rate or direction. Rawlsian fairness requires that procedures be chosen from behind a "veil of ignorance" about which position one will occupy; the moving-knife protocol embodies this principle, since neither party knows in advance whether they will be the caller or the recipient.

Puppe and Tasnadi 2026 adapt the moving-knife framework to redistricting by reinterpreting "the resource" as the state's geographic territory and "the knife" as a directional sweep line. Instead of moving across the state at a fixed rate, the knife rotates through angular orientations. At each orientation, the "stop" criterion is replaced by the population balance condition: the split point is wherever the cumulative population first reaches $P_{\text{total}}/2$. The "value" being optimised is Reock compactness: the algorithm stops not when either party calls stop, but at the orientation that maximises the geometric quality of the split.

2.2. Reock Compactness

Reock compactness was introduced by Ernest Reock in 1961 as a "measure of the shape of geographic districts" [Reock, 1961]. The measure is defined for a district D as:

$$\text{Reock}(D) = \frac{A(D)}{A(\text{MEC}(D))}, \tag{1}$$

where $A(D)$ is the area of district D and $A(\text{MEC}(D))$ is the area of the smallest circle that encloses D (the minimum enclosing circle, or MEC). Since $D \subseteq \text{MEC}(D)$ by definition, we have $\text{Reock}(D) \in (0, 1]$, with equality iff D is a perfect circle.

Remark 1 (Centroid approximation)**.** *The precise Reock score of a district requires the MEC of the district's full polygon boundary. In the Phase 1 implementation described in this paper, we compute the MEC of the projected tract centroid points only (not the full polygon vertices). This approximation may yield* $\text{Reock}_{\text{approx}}(D) > 1.0$ *for districts that contain large-area boundary tracts whose polygon vertices extend beyond the centroid-MEC. The implementation clamps all Reock scores to* $[0.0, 1.0]$*:* $\text{Reock}_{\text{clamped}} = \min(\text{Reock}_{\text{approx}}, 1.0)$*. The invariant* $\text{Reock} \leq 1$ *is enforced by clamping, not by structural guarantee of the centroid approximation. Phase 2 will implement the polygon-boundary MEC.*

Legal prominence. Reock is the most widely cited compactness measure in redistricting litigation for two reasons. First, it is intuitively interpretable: a district with $\text{Reock} = 0.40$ uses 40 % of the smallest circle that could contain it. Second, elongation and protrusion —

the most common forms of partisan packing and cracking — directly reduce Reock, because elongated shapes have large MECs relative to their area. Polsby-Popper is sensitive to perimeter irregularity ("jagged" edges from following county or precinct boundaries) even in compact districts; Reock is not, because it depends only on the overall extent of the district, not its boundary smoothness.

2.3. Welzl's Minimum-Enclosing-Circle Algorithm

Given a set of m points in the plane, Welzl's algorithm [Welzl, 1991] computes the smallest enclosing circle in expected $O(m)$ time via a randomised incremental procedure. The algorithm maintains a candidate MEC and processes points one at a time. If a new point lies outside the current MEC, the point is added to the *basis* (the boundary-defining set) and the MEC is recomputed from the updated basis. In the redistricting context, the input points are the projected centroids of the tracts in the half being evaluated; the basis has at most three points (the MEC of a set of 2D points has at most 3 boundary points).

Proposition 1 (Welzl Runtime [Welzl, 1991])**.** *Welzl's algorithm computes the MEC of m points in $\mathbb{R}^2$ in expected $O(m)$ time.*

The expected-$O(m)$ bound makes Welzl's algorithm well-suited to the inner loop of MKA: it is called $2 \times n_{\text{orient}}$ times per bisection node (once for each half at each orientation), contributing $O(n_{\text{orient}} \cdot m \log m)$ total work (the $O(m \log m)$ sort step dominates).

2.4. Relationship to Existing Platform Algorithms

Table 1 places MKA in the context of the four main structure algorithms in the redistricting platform.

MKA as the third geometric algorithm. CVD-Geo (B.22b) and BfsGrowth (B.23 [Della-Libera, 2026a]) each introduced a new geometric structure algorithm targeting a distinct court-relevant metric: CVD-Geo targets Polsby-Popper via geographic proximity clustering; BfsGrowth targets geographic coverage balance. MKA is the third, targeting Reock via angular sweep. The three algorithms form a family: each uses Albers EPSG:5070 projection for geographic coordinates, each shares the 200-iteration boundary-swap rebalance, and each provides a distinct plain-language justification for the submitted plan.

Table 1: Structure algorithm comparison. MKA is the only algorithm that directly targets Reock; the others optimise different metrics and achieve Reock as a secondary consequence. Runtime is per bisection node with m tracts.

Algorithm	Optimises	Distance	Reock	Runtime
Metis	Edge cuts	Graph topology	Low	$O(m \log m)$
CVD	Proximity	BFS hop-count	Medium	$O(n_{\text{iter}} \times m)$
CVD-Geo	Proximity	Euclidean	Medium–high	$O(n_{\text{iter}} \times m)$
BfsGrowth	Balance speed	Euclidean seeds	Medium	$O(m \log m)$
MKA	**Reock**	**Euclidean sweep**	**Highest**	$O(n_{\text{orient}} \cdot m \log m)$

Why not optimise Polsby-Popper in MKA? Polsby-Popper requires perimeter computation, which in turn requires the polygon boundary data from TIGER/Line shapefiles. Reock requires only the positions of tract centroids and the MEC geometry, both available from the existing centroid companion file. MKA therefore defaults to Reock; an optional `-mka-metric polsby` flag is supported for contexts where Polsby-Popper is the legally operative metric, but Reock is the preferred default because it requires no polygon boundary data beyond what CVD-Geo already uses.

3. The Moving-Knife Algorithm

3.1. Notation

Let $G_v = (V_v, E_v)$ be the subgraph at bisection tree node v, with $m = |V_v|$ tracts and population p_i for each tract $i \in V_v$. Let $\ell_i = (\lambda_i, \phi_i)$ be the WGS84 (longitude, latitude) of the population-weighted centroid of tract i's polygon, loaded from the centroid companion file (Section 3.6). Let $\mathbf{x}_i = (x_i, y_i) = \text{proj}(\ell_i)$ be the Albers-projected coordinates in metres (EPSG:5070, shared with CVD-Geo [Della-Libera, 2026c]).

3.2. Design Choices

Three non-obvious design decisions underlie the algorithm.

Choice 1: $n_{\text{orient}} = 180$ (every $1°$). The sweep tests orientations $\theta \in \{0, 1, \ldots, 179\}$ degrees, covering $[0°, 180°)$ uniformly. Restricting to $[0°, 180°)$ is correct because opposite orientations produce the same bisecting hyperplane with left and right swapped; no information is lost by this restriction. The 1° granularity ensures the globally best direction is never missed by more than half a degree. Runtime is $O(n_{\text{orient}} \cdot m \log m) \approx O(180m \log m)$ per bisection node, which is comparable to a single iteration of CVD-Geo and substantially faster than the full CVD-Geo run (typically 7–12 iterations $\times$ $O(m)$).

Choice 2: Reock as the default metric. The sweep scores each split by $\min(\text{Reock}_{\text{left}}, \text{Reock}_{\text{right}})$: the worst-case Reock score of the two halves. Using the minimum (rather than the mean) ensures both halves are compact, following the maximin principle: maximise the outcome for the worst-off district. Reock is the default (rather than Polsby-Popper) because Reock requires only centroid positions and MEC geometry; Polsby-Popper requires polygon perimeter data, introducing an additional data dependency. The flag `-mka-metric polsby` enables Polsby-Popper scoring for jurisdictions where that metric is legally operative.

Choice 3: Post-hoc rebalance. The linear sweep produces a population split that is close to but not exactly balanced: the binary search for the split point may overshoot by at most one tract's population. The same 200-iteration boundary-swap rebalance used by CVD-Geo and BfsGrowth enforces exact balance within tolerance. The rebalance also repairs contiguity: for non-convex subgraph geometries, a linear sweep may produce disconnected half-subgraphs; BFS-based contiguity repair is applied before returning the final split.

3.3. Seed Derivation

MKA is deterministic given `base_seed` and n_{orient}. A seed is needed only for tie-breaking when two orientations produce identical minimum-Reock scores (rare in practice).

Definition 1 (MKA Node Seed)**.** *Let* `base_seed` *be the unsigned 64-bit global seed and* `node_path` *the path string for bisection tree node v. The* MKA node seed *is:*

$$s_v^{\text{mka}} = \text{first8}\Big(\text{SHA-256}(\texttt{"MKA\_INIT\_"} \parallel \text{le4}(|\texttt{node\_path}|) \parallel \texttt{node\_path} \parallel \text{le8}(\texttt{base\_seed}))\Big),$$

where $\text{le4}(n)$ *encodes n as a 4-byte little-endian* `u32`*,* $\text{le8}(x)$ *encodes x as an 8-byte little-endian* `u64`*, and* $\text{first8}(h)$ *reads the first 8 bytes of the SHA-256 digest as a little-endian* `u64` *[National*

Institute of Standards and Technology, 2015].

The length prefix le4(|`node_path`|) eliminates prefix-collision risk: the boundary between `node_path` and `base_seed` is unambiguous. The prefix `"MKA_INIT_"` is distinct from `"CVD_GEO_INIT_"` (Phase 2 CVD-Geo) and `"CVD_INIT_"` (Phase 1 CVD). A platform invariant test asserts all three prefix constants are present in source and are pairwise distinct.

3.4. Formal Pseudocode

Algorithm 1 gives the complete MKA bisection procedure.

The **Reock** subroutine (line 17) computes:

$$\mathrm{Reock}(S, \mathbf{x}, A) = \min\left(1.0,\ \frac{\sum_{i \in S} A_i}{\pi \cdot r_{\mathrm{MEC}}(S, \mathbf{x})^2}\right),$$

where A_i is the TIGER/Line polygon area of tract i in square metres, and $r_{\mathrm{MEC}}(S, \mathbf{x})$ is the radius of the smallest enclosing circle of the projected centroid points $\{\mathbf{x}_i : i \in S\}$, computed by Welzl's algorithm (Proposition 1). The clamp enforces the Reock ≤ 1 invariant under the centroid approximation (Remark 1).

3.5. Runtime Complexity

Proposition 2 (MKA Runtime)**.** *MKA bisection at a node with m tracts requires:*

- *Per orientation (n_{orient} total):*
  - *Projection: $O(m)$ dot products.*
  - *Sort: $O(m \log m)$ (dominated by projection if m is small; note that the sort can be replaced by a linear scan when n_{orient} is large, since successive orientations produce nearly-sorted projections).*
  - *Population binary search: $O(m)$ cumulative sum.*
  - *Welzl MEC (two calls): expected $O(m)$ each.*
- *Post-hoc rebalance: $O(200m)$.*
- *Total: $O(n_{\mathrm{orient}} \cdot m \log m + 200m) = O(n_{\mathrm{orient}} \cdot m \log m)$.*

For $n_{\text{orient}} = 180$*:* $O(180m \log m)$*. This is comparable to one full CVD-Geo iteration count (typically 7–12)* $\times$ $O(m)$ *per iteration* $= O(84m)$*–*$O(144m)$*. The sort step dominates; using a radix sort reduces the per-orientation cost to* $O(m)$*, giving total* $O(180m)$ *— strictly faster than CVD-Geo.*

3.6. Data Requirements

MKA requires the same centroid companion file introduced by CVD-Geo: `{state}_centroids_{year}.js` stored alongside the adjacency file in `data/{year}/`. It additionally requires the tract area field (`LoadedGraph.tract_areas`), which stores TIGER/Line polygon areas in square metres and is bundled with the existing adjacency file. No additional fetch step is needed if centroids were already fetched for CVD-Geo Phase 2.

If the centroid companion file is absent and `-structure moving-knife` is requested, the CLI returns a diagnostic error:

```
ERROR: --structure moving-knife requires tract centroid data.
       Run: bisect fetch --type centroids --states NC --year 2020
```

No silent fallback to a different algorithm occurs.

3.7. CLI and YAML

Pure MKA structure:

```
bisect state --state NC --year 2020 \
  --structure moving-knife \
  --mka-orientations 180 \
  --mka-metric reock
```

Hybrid: MKA direction + AreaSection balance enforcement: The `split_subgraph_mka_directi` function returns the optimal angle $\theta^* \in [0, \pi)$ without performing the full split. AreaSection uses θ^* to override its `ratio_direction` before applying its own balance logic:

```
bisect state --state NC --year 2020 \
  --structure area-section \
  --area-section-init moving-knife
```

YAML configuration:

```
algorithm:
  structure: moving-knife
  mka_orientations: 180
  mka_metric: reock           # or: polsby-popper
  weights: geographic
  search: convergence
  convergence_threshold: 600
  balance_tolerance: 0.5
workers: 8
```

3.8. Audit Chain

Every MKA run appends the following fields to `runs/{label}/{year}/index.json`:


```
"structure":          "moving-knife",
"mka_orientations":   180,
"mka_metric":         "reock",
"mka_seed_formula":   "SHA-256('MKA_INIT_'||node_path_len:u32le
                        ||node_path||base_seed:u64le)",
"node_path":          "01",
"base_seed":          12345678,
"mka_seed":           987654321,
"optimal_angle_deg": 47.0,
"reock_left":         0.412,
"reock_right":        0.389,
"min_reock_score":    0.389,
"convergence_warning": false
```


`optimal_angle_deg` enables independent verification: an auditor reruns the sweep with the same centroids and n_{orient} and confirms θ^* matches. `node_path` is required for independent seed reproduction. `convergence_warning` is always `false` for MKA: the sweep is exhaustive over all orientations and has no iterative convergence criterion. `min_reock_score` is the quantity MKA directly maximises.

4. Empirical Comparison

4.1. Experimental Setup

We compare four structure algorithms on three states representing the spectrum from regular geometry to geometrically challenging:

- **NC** (North Carolina, $k = 14$, $n \approx 2{,}660$ tracts): compact rectangular geometry, competitive partisan state. Reock scores are already moderate for METIS due to the regular shape. The FL panhandle geometry is absent, so the orientation sweep is expected to provide a modest but measurable gain.

- **FL** (Florida, $k = 28$, $n \approx 4{,}200$ tracts): the 200-mile panhandle extends westward from the main peninsula; coastal geometry around Tampa Bay, Miami, and the Keys creates elongated and concave subgraph regions. Gains from angle-optimised splits are expected to be largest here, because naive horizontal or vertical sweeps miss the locally optimal orientation for panhandle subgraphs.

- **WA** (Washington, $k = 10$, $n \approx 1{,}400$ tracts): Puget Sound creates a geographic discontinuity between the westside (Kitsap Peninsula, Olympic Peninsula) and the eastside (Cascade foothills, eastern plains). Intermediate geometry between NC and FL.

All runs use 2020 Census data, geographic boundary-length edge weights (B.2 default), standard bisection tree, and base seed $s = 42$. Single-run results are marked with † (not ensemble-averaged); the purpose of this table is to illustrate *relative ordering* of Reock performance, not to report production-quality compactness figures. Production figures from *G.15 Comprehensive Comparison* [Della-Libera, 2026b] should be used for plan certification.

Algorithms.

- **Metis**: standard METIS graph partitioning, `-structure standard-bisect` (the platform default).

- **CVD-Geo**: Geographic CVD Phase 2, `-structure centroidal-voronoi -cvd-metric geographic`, 20 iterations [Della-Libera, 2026c].

- **BfsGrowth**: BFS Growth, `-structure bfs-growth` [Della-Libera, 2026a].
- **MKA**: Moving-Knife Algorithm, `-structure moving-knife -mka-orientations 180 -mka-metric reock`.

Metrics.

- **Min-Reock**: minimum Reock score across all k districts; this is the quantity MKA directly maximises. Higher is better.
- **Mean-Reock**: mean Reock score across all k districts. Higher is better.
- **EC**: total weighted edge cut of the final plan. Lower is better.
- **Time**: wall-clock time in seconds on a reference workstation (8 cores, `-workers 8`).

4.2. Results

Table 2 presents the Reock comparison.

4.3. Analysis

Min-Reock improvement. MKA achieves the highest minimum Reock score on all three states. Over the next-best algorithm (BfsGrowth in every case), the improvement is: NC: +0.019 (+8.5 %); FL: +0.025 (+13.8 %); WA: +0.024 (+11.8 %). The improvement is largest on FL, consistent with the hypothesis that the panhandle geometry provides large gains when the optimal orientation is found rather than defaulting to a fixed radial growth or graph-topology partition. NC shows the smallest gain, because the regular geometry of the Piedmont plateau means nearly all orientations produce similar splits — the sweep finds a modest improvement but the feasible range is narrower.

Mean-Reock. Mean Reock follows the same ordering. The mean–min gap is approximately 0.03–0.05 for all algorithms, indicating that Reock is reasonably uniform across districts for all methods; no algorithm produces one extraordinarily compact district at the expense of others.

Table 2: Reock compactness and edge-cut comparison on NC ($k = 14$), FL ($k = 28$), and WA ($k = 10$), 2020 census, base seed $s = 42$. †: single run; not ensemble-averaged. Min-Reock: minimum Reock across all k districts (higher is better). EC: total edge cut (lower is better). Bold: best Min-Reock per state.

State	Algorithm	Min-Reock†	Mean-Reock†	EC†	Time (s)
NC ($k = 14$)	Metis	0.187	0.241	4,821	4.2
	CVD-Geo	0.219	0.268	5,512	5.9
	BfsGrowth	0.224	0.271	6,104	3.8
	MKA	**0.243**	**0.279**	6,318	6.1
FL ($k = 28$)	Metis	0.152	0.201	9,247	9.1
	CVD-Geo	0.174	0.228	10,219	10.7
	BfsGrowth	0.181	0.233	11,037	7.4
	MKA	**0.206**	**0.251**	11,482	11.3
WA ($k = 10$)	Metis	0.171	0.219	3,614	3.3
	CVD-Geo	0.198	0.247	3,971	4.0
	BfsGrowth	0.203	0.251	4,318	2.9
	MKA	**0.227**	**0.263**	4,591	4.3

Edge cut. MKA has the highest edge cut on all three states, as expected: it does not minimise edge cuts (that is Metis's objective) and its angular sweep does not align with the graph-topology boundaries that METIS exploits. The excess EC over Metis is 31 % on NC, 24 % on FL, and 27 % on WA. The excess over CVD-Geo is 15 % on NC, 12 % on FL, and 16 % on WA. These are acceptable for most redistricting contexts: EC is a secondary criterion when the primary mandate is Reock-based compactness; and the 10–15 % EC premium over CVD-Geo translates to modestly less boundary-following districts, not to grossly non-contiguous ones.

Comparison with prior geometric algorithms. CVD-Geo improves Polsby-Popper over Metis by approximately 8–13 % (B.22b [Della-Libera, 2026c]). MKA improves Min-Reock over Metis by approximately 30 % on NC, 36 % on FL, and 33 % on WA — substantially larger gains, because the sweep directly targets the Reock geometry. Each algorithm is best-in-class for its own objective.

4.4. Orientation Sensitivity

To characterise how strongly performance depends on orientation granularity, Table 3 compares $n_{\text{orient}} \in \{36, 90, 180\}$ on NC ($k = 14$).

Table 3: MKA orientation granularity sensitivity: NC ($k = 14$), 2020, seed $s = 42$. †: single run. Diminishing returns beyond 90 orientations on NC.

n_{orient}	Min-Reock†	EC†	Time (s)
36	0.237	6,291	1.4
90	0.241	6,309	3.2
180	0.243	6,318	6.1

The 36-orientation sweep captures approximately 97.5 % of the 180-orientation Min-Reock at 23 % of the runtime. The default $n_{\text{orient}} = 180$ is recommended for production use to ensure the globally best direction is not missed by more than 0.5°. For rapid exploratory runs, `-mka-orientations 36` reduces runtime to approximately 1.4 s on NC. Phase 2 will benchmark 36 vs. 180 on FL and WA to confirm whether the 5× speedup justifies the small quality loss in panhandle geometries.

4.5. Partisan Neutrality Check

Table 4 verifies that the Reock improvement does not introduce systematic partisan bias.

State	k	Metis D-seats	CVD-Geo D-seats	MKA D-seats
NC	14	7	7	7
FL	28	12	12	12
WA	10	6	6	6

Table 4: 2020 presidential two-party Democratic seat share under Metis, CVD-Geo, and MKA (seed $s = 42$). †: single run. The angular sweep selects the most compact orientation; it does not select for or against either party's seat share.

Partisan outcomes are identical across all three algorithms on all three states at seed $s = 42$. The MKA angular sweep is parametrised by geographic coordinates and population balance; it contains no electoral data and has no mechanism to advantage either party. The partisan-neutrality result is consistent with the fair-division theoretical justification

(Section 5). Ensemble verification across 5 independent seeds confirms stable D-seat outcomes; we defer to the G.15 comprehensive comparison for multi-seed partisan analysis.

5. The Fair-Division Legal Argument

5.1. The Problem of Algorithmic Justification

When a redistricting commission submits a plan to a court, it must provide a justification: why is this plan the best available, and why is the procedure that produced it neutral? Algorithmic redistricting provides mathematically precise answers to both questions, but the precision creates a new challenge: the mathematical arguments are often inaccessible to non-technical judges and lay jurors.

Consider the METIS justification: "we minimised the total edge-weighted boundary length of the bisection tree, where edge weights are proportional to the shared geographic boundary length of adjacent census tracts, using the multilevel Kernighan-Lin algorithm." This is a correct and precise description of what METIS does. But it requires the listener to understand graph partitioning, weighted adjacency, and multilevel coarsening to evaluate whether it is a good criterion for compactness. In *Common Cause v. Lewis*, the North Carolina Superior Court found it necessary to appoint a special master to evaluate the technical properties of proposed maps — a symptom of the accessibility gap between algorithmic precision and legal comprehension.

5.2. The Moving-Knife Certificate

The moving-knife protocol provides an alternative justification that is both mathematically precise and intuitively accessible:

> *A neutral sweep line was rotated through 180 orientations across the state. At each orientation, the line was positioned where it would divide the state's population equally. The orientation that produced the most geometrically compact pair of regions — as measured by the Reock score, the ratio of each region's area to the smallest circle that could contain it — was selected. Neither party chose the orientation, the line position, or the compactness criterion: all three were fixed by the algorithm before any district boundaries were drawn.*

This description is complete and accurate. It is also understandable by a non-technical reader: the intuition of a "neutral sweep" is familiar from everyday geometric reasoning, and Reock compactness ("what fraction of the smallest enclosing circle does the district fill?") is more concrete than "minimise total edge-weighted boundary length."

5.3. Rawlsian Procedural Fairness

The philosophical foundation for the moving-knife certificate is Rawlsian procedural fairness. Rawls 1971 argued that a distribution is just if it could have been chosen by rational agents behind a "veil of ignorance" — that is, without knowing which position in the distribution they would occupy. For redistricting, this translates to: a plan is procedurally fair if the procedure that produced it would have been acceptable to either party before knowing which districts they would be assigned.

The moving-knife protocol satisfies this criterion. Before the sweep begins, neither the commission nor any party knows which half of the state will contain which group of voters — the partition depends on the geographic orientation that maximises Reock, which is a function of tract geometry and population distribution, not of political affiliation. The two halves of the split are not labelled "Republican" or "Democratic" during the algorithm's execution; they are labelled only after the plan is complete.

This stands in contrast to any procedure that selects among candidate plans based on partisan criteria, which would fail the Rawlsian test because such a procedure could only be acceptable to the advantaged party.

Puppe and Tasnadi 2026 formalise this argument in the public-choice setting, proving that a moving-knife redistricting procedure satisfying anonymity (no party is identified in the algorithm), population balance (both halves receive equal population), and compactness maximisation (the most compact orientation is selected) is the unique procedure satisfying all three axioms simultaneously. Their axiomatic result provides academic backing for the fair-division certificate.

5.4. Comparison to the METIS Argument

The Metis argument is also neutral: minimising edge-weighted boundary length is a function of geography only, containing no electoral data. Both arguments are procedurally impartial in the sense that neither party controls the outcome. We argue that the MKA argument is preferable in three contexts:

1. **Jurisdictions where Reock is the operative metric.** When a state constitution or court order specifies Reock compactness as the required criterion, MKA directly satisfies the legal mandate. METIS does not: it minimises edge cut, not Reock, and cannot certify that the submitted plan maximises Reock.

2. **Courts that prioritise procedural intelligibility.** When a court must evaluate the procedure on its own merits — not just the output — the moving-knife sweep is more accessible than the multilevel coarsening of METIS. The sweep has a direct physical analogy (a ruler rotating across a map) that a non-technical judge can visualise; multilevel Kernighan-Lin does not.

3. **Challenges alleging manipulation.** When a plaintiff alleges that the algorithm was configured to favour one party, the moving-knife procedure is more resistant to this attack: the only configurable parameters are the number of orientations (180 by default) and the compactness metric (Reock by default), both of which are publicly specified and audit-logged. The audit chain records `optimal_angle_deg`, allowing an independent auditor to rerun the sweep and confirm that the certified angle is correct.

5.5. Limitations of the Fair-Division Argument

The moving-knife certificate is not a complete defence in all redistricting contexts. Several limitations apply:

VRA majority-minority requirements. Section 5 of the Voting Rights Act and state analogues may require that minority-majority districts be preserved. MKA does not account for racial composition; a plan that maximises Reock may or may not satisfy VRA requirements. Practitioners must verify VRA compliance using the platform's VRA analysis tools (`bisect analyze -types vra`) separately from the Reock compactness certification.

Multi-criterion compactness standards. Some state constitutions require compactness under multiple metrics. Florida's Article III, §20, has been interpreted to require consideration of both Reock and Polsby-Popper [Florida Legislature, 2010]. A plan certified as Reock-optimal by MKA may not be Polsby-Popper-optimal. In such jurisdictions, the `-mka-metric polsby` flag or a multi-run comparison with CVD-Geo may be appropriate.

Contiguity and non-convexity. For non-convex subgraph geometries (peninsulas, panhandles), a linear sweep may not produce the globally Reock-optimal split: the best Reock orientation may depend on the boundary shape of the subgraph in a way that a projection sweep cannot capture. MKA is a best-available heuristic for non-convex regions, not a global optimiser. The audit chain records `optimal_angle_deg` to allow verification that the sweep was exhaustive at the reported granularity, but does not certify global optimality.

Bisection tree structure. MKA applies the moving-knife sweep at each bisection node independently. The globally Reock-optimal plan may not be achievable by independent local Reock maximisation at each node, since node-level decisions interact. This is the same limitation that applies to all recursive bisection algorithms.

6. Conclusion

We have introduced the Moving-Knife Algorithm (MKA), the first redistricting structure algorithm that explicitly maximises Reock compactness. MKA adapts the fair-division moving-knife protocol of Dubins and Spanier 1961 to the bisection framework, sweeping through 180 orientations and selecting the one that maximises the minimum Reock score of the two population-balanced halves. Implementation reuses the Albers EPSG:5070 projection and centroid companion file from CVD-Geo (B.22b [Della-Libera, 2026c]), and the 200-iteration boundary-swap rebalance shared with all geometric structure algorithms.

When to use MKA. Use `-structure moving-knife` when:

- The jurisdiction's constitutional or statutory standard explicitly cites Reock compactness (North Carolina, Florida, Wisconsin — see Section 1).
- A court order or remedial mandate specifies that the submitted plan must demonstrate Reock maximisation.

- A procedurally neutral, legally articulable justification is needed that is more accessible to non-technical judges than the graph-theoretic arguments of Metis or the Lloyd-iteration arguments of CVD-Geo.

When other algorithms may be preferable.

- For Polsby-Popper optimisation: use CVD-Geo (`-structure centroidal-voronoi -cvd-metric geographic`).
- For minimum edge cut: use Metis (`-structure standard-bisect`).
- For geographic balance speed: use BfsGrowth (`-structure bfs-growth`).
- When Reock is not the operative metric: MKA's 10–15 % EC premium over CVD-Geo may not be justified.

Position in the algorithm portfolio. MKA joins CVD-Geo (B.22b) and BfsGrowth (B.23 [Della-Libera, 2026a]) as the third geometric structure algorithm in the platform. Together, the three algorithms provide coverage of the three compactness metrics most commonly cited in redistricting litigation: BfsGrowth targets geographic balance (used as a proxy for Convex Hull Ratio), CVD-Geo targets Polsby-Popper via geographic proximity, and MKA targets Reock via angular sweep. Practitioners selecting among them should match the algorithm to the legally operative metric in their jurisdiction.

Open questions. Several extensions are deferred:

1. **Polygon-boundary MEC (Phase 2)**: replace the centroid approximation (Remark 1) with the full polygon-boundary MEC using the TIGER/Line shapefile vertex data. This will eliminate the Reock > 1 clamping cases and produce exact Reock scores.
2. n_{orient} **tradeoff**: benchmark 36 vs. 180 orientations on FL and WA panhandle geometries to determine whether the $5\times$ speedup of 36 orientations is worth the small quality loss.
3. **MKA + BisectionEnsemble**: run BisectionEnsemble($T = 50$) with MKA bisectors to find the best Reock-maximising seed among 50 independent runs. Per-run runtime ($\approx$6 s on NC) makes the ensemble feasible with `-workers 8`.

4. **Non-convex subgraphs**: for states with severe non-convexity (peninsulas, island tracts), investigate whether a post-hoc local search over orientations — seeded by MKA's θ^* — can improve on the linear sweep's heuristic.

Summary. The Moving-Knife Algorithm provides a principled, court-defensible procedure for Reock maximisation in congressional redistricting. Its theoretical basis in fair-division theory (Dubins-Spanier 1961, Puppe-Tasnadi 2026) provides academic grounding; its empirical performance (highest Min-Reock on NC, FL, and WA) confirms the operational claim; and its fair-division certificate offers a procedurally neutral justification that is accessible to non-technical decision-makers. The algorithm is available in the `bisect` platform as `-structure moving-knife`.

References

Common Cause v. Lewis. No. 18 cvs 014001 (N.C. superior court, 2019), 2019. North Carolina Superior Court decision striking down the 2017 legislative maps as unconstitutional partisan gerrymanders. Expert witnesses presented Reock scores for each district; the court's remedial order referenced Reock as a compactness benchmark. Establishes Reock as a judicially cognisable compactness metric in North Carolina redistricting litigation.

Gio Della-Libera. BfsGrowth: Geographic balance via radial seed expansion, 2026a. B.23 in Algorithmic Redistricting series. Introduces BfsGrowth, which grows districts outward from geographically dispersed seeds, targeting geographic balance speed. B.25 uses BfsGrowth as one of the four comparison algorithms in the empirical Reock evaluation; BfsGrowth consistently places second-best on Min-Reock, behind MKA.

Gio Della-Libera. Ensemble comparison of structure-layer algorithms for congressional redistricting, 2026b. G.15 in Algorithmic Redistricting series. Comprehensive 50-state ensemble comparison providing definitive algorithm ranking. B.25 single-run results are complemented by G.15 ensemble figures for production use.

Gio Della-Libera. Geographic centroidal Voronoi districts: True Euclidean proximity via Albers projection, 2026c. B.22b in Algorithmic Redistricting series. Introduces CVD-Geo, Phase 2 CVD using Euclidean distance on EPSG:5070 Albers-projected tract centroids. B.25

MKA reuses the Albers projection, centroid companion file format, and boundary-swap rebalance from B.22b without modification.

Lester E. Dubins and Edwin H. Spanier. How to cut a cake fairly. *American Mathematical Monthly*, 68(1):1–17, 1961. doi: 10.2307/2311357. Introduces the moving-knife protocol for fair division of a heterogeneous resource among n parties. Proves proportionality and envy-freeness of the continuous knife-sweep procedure. B.25 adapts this protocol to redistricting by replacing the linear knife sweep with an angular orientation sweep, evaluated for Reock compactness at each orientation.

Florida Legislature. Florida constitution, article III, section 20: Standards for legislative apportionment, 2010. Florida's constitutional compactness standard, enacted via Amendment 6 (2010). Florida courts have applied Reock and convex-hull ratio alongside Polsby-Popper in redistricting litigation. Used in B.25 as exemplar of a multi-metric compactness jurisdiction where MKA satisfies the Reock component of the standard.

National Institute of Standards and Technology. Secure hash standard (SHS). Technical Report FIPS 180-4, NIST, 2015. URL `https://doi.org/10.6028/NIST.FIPS.180-4`. SHA-256 specification used in the MKA node seed derivation. The domain-separation prefix `"MKA_INIT_"` is distinct from `"CVD_GEO_INIT_"` and `"CVD_INIT_"`, ensuring independence across all structure-layer seeds in the audit chain.

Clemens Puppe and Attila Tasnadi. Moving-knife redistricting. *Public Choice*, 2026. Formalises the moving-knife protocol for redistricting. Proves that any procedure satisfying anonymity, population balance, and compactness maximisation is uniquely characterised by a moving-knife angular sweep. B.25 is the computational implementation and empirical evaluation of the Puppe-Tasnadi proposal.

John Rawls. *A Theory of Justice*. Harvard University Press, 1971. Foundational text for Rawlsian procedural fairness and the "veil of ignorance" argument. B.25 invokes the Rawlsian framework to argue that the moving-knife protocol is procedurally just: the orientation and split point are determined by geometry alone, without reference to which party will occupy which half.

Ernest C. Reock. A note: Measuring compactness as a requirement of legislative apportionment. *Midwest Journal of Political Science*, 5(1):70–74, 1961. doi: 10.2307/2109043.

Introduces Reock compactness: the ratio of district area to the area of the minimum enclosing circle. The measure has become the most frequently cited compactness metric in redistricting litigation in the United States, cited by courts in North Carolina, Wisconsin, Florida, and Georgia. B.25 MKA directly maximises this measure.

Emo Welzl. Smallest enclosing disks (balls and ellipsoids). In Hermann Maurer, editor, *New Results and New Trends in Computer Science*, volume 555 of *Lecture Notes in Computer Science*, pages 359–370. Springer, 1991. doi: 10.1007/BFb0038202. Presents the randomised incremental algorithm for computing the minimum enclosing circle of a set of m points in expected $O(m)$ time. Used in B.25 MKA to compute Reock scores at each orientation of the sweep. The expected-$O(m)$ bound is essential to keeping the total sweep runtime at $O(n_{\text{orient}} \times m)$.

Algorithm 1 MKA-BISECT(G_v, $\mathbf{x}$, $\mathbf{p}$, A, n_{orient}, `base_seed`, `node_path`, `balance_tolerance`)

$m \leftarrow |V_v|$; $P \leftarrow \sum_{i \in V_v} p_i$
if $m < 2$ **then return** degenerate assignment
end if
$s_v \leftarrow$ MKA-Seed(`base_seed`, `node_path`) ▷ Definition 1
`sorted` $\leftarrow$ sort(V_v) ▷ sort by global tract index
$\mathbf{x}' \leftarrow [\mathbf{x}_i : i \in$ `sorted`$]$ ▷ projected coordinates in sorted order
`best_score` $\leftarrow -\infty$; `best_left` $\leftarrow \emptyset$; `best_right` $\leftarrow \emptyset$
for $k = 0$ **to** $n_{\text{orient}} - 1$ **do**
    $\theta \leftarrow k \cdot \pi / n_{\text{orient}}$ ▷ angle in $[0, \pi)$
    $\mathbf{u} \leftarrow (\cos\theta, \sin\theta)$ ▷ unit sweep vector
    $\text{proj}_i \leftarrow \mathbf{x}'_i \cdot \mathbf{u}$ for all i ▷ project onto sweep axis
    $\sigma \leftarrow$ sort indices by proj_i ascending
    $C \leftarrow 0$; `split` $\leftarrow 0$
    for $j = 0$ **to** $m - 1$ **do**
        $C \leftarrow C + p_{\sigma[j]}$
        if $C \geq P/2$ **then** `split` $\leftarrow j$; **break**
        end if
    end for
    $L \leftarrow \{$`sorted`$[\sigma[j]] : j \leq$ `split`$\}$
    $R \leftarrow \{$`sorted`$[\sigma[j]] : j >$ `split`$\}$
    $r_L \leftarrow$ Reock($L, \mathbf{x}, A$) ▷ Reock via Welzl MEC; clamped to $[0, 1]$
    $r_R \leftarrow$ Reock($R, \mathbf{x}, A$)
    `score` $\leftarrow \min(r_L, r_R)$
    if `score` $>$ `best_score` **then**
        `best_score` $\leftarrow$ `score`; `best_left` $\leftarrow L$; `best_right` $\leftarrow R$
    else if `score` $=$ `best_score` **then**
        tie-break by s_v
    end if
end for
(`best_left`, `best_right`) $\leftarrow$ Rebalance(`best_left`, `best_right`, G_v, $\mathbf{p}$, `balance_tolerance`) ▷ 200-iter boundary-swap
return (`best_left`, `best_right`)